

The company has not named any partners for BH-03, but Tesla Motors, BMW and Volkswagen may be interested in the project. It also has not given a release date for the two new tires.

Tata Jaguar Land Rover to build first electric car

Tata Motors-owned Jaguar Land Rover (JLR) is set to build its first all-electric car in Austria. The luxury car-maker's Jaguar brand is expected to begin production next year. This car would be the latest in a string of launches after the new XE baby Jag and F-Pace sports utility vehicle.

The company is believed to be devising plans for a small production line in Austria, making about 10,000 cars a year. Jaguar wants to build an electric vehicle, in part, to sidestep looming European Union emission fines.

Tata Group had paid US\$ 2.3 billion in 2008 to buy JLR from Ford and has more than doubled its workforce to 32,000. The company now sells almost half-a-million cars, and is also planning to build a factory in America.

Finger-mounted device converts text to audio for the visually impaired

Researchers from MIT have developed a finger-worn device with a built-in camera that converts written text into audio for the visually impaired. The device provides feedback, tactile and audible, which guides the user's finger along a line of text, and the system generates the corresponding audio in real time.

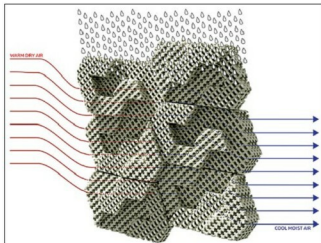
Roy Shilkrot, an MIT graduate in media arts and sciences, and his colleagues tested many variations of their device in a study with vision-impaired volunteers. One included two haptic motors, one on top of the finger and the other beneath it. The vibration of motors indicated whether the subject should raise or lower the tracking finger.

Another version, without motors, used audio feedback—a musical tone that increased in volume if the user's finger began to move away from the line of text. Researchers also tested the motors and the musical tone in conjunction.

However, they were unable to get consensus on which type of feedback was most useful. They are now concentrating on audio feedback, since it allows for a smaller, lighter-weight sensor. The key to the device's performance is an algorithm for processing the camera's video feed, which Shilkrot and his colleagues developed.

3D-printed brick that can cool your house

For centuries, people living in hot, dry places have harnessed the natural phenomenon of evaporating water to cool their homes, often through something called the Muscatese Evaporative system.



Cool Brick, the 3D-printed ceramic brick

In this low tech set-up, a wooden lattice covers the windows while a porous, water-filled ceramic jar sits inside it. As air passes through the lattice, water inside the jar evaporates, depositing a stream of cool air into the home.

The passive cooling system is inexpensive and energy-efficient and happens to be quite relevant today, as proven by 3D printing company, Emerging Objects.

The Oakland based company has translated the phenomenon of evaporative cooling into a 3D-printed ceramic brick named Cool Brick. This hand-sized piece of ceramic is essentially a hybrid of a wooden lattice and ceramic jar rolled into a single, lightweight slab.

According to its designers, the porous material can soak up water like a sponge; so when air passes through it, much like a wooden lattice, the water held in the micropores evaporates and introduces a cooling effect.

Faster paper-thin, flexi-screens for computers

The next generation of transistors may pave the way for flexible, paper-thin computer screens that provide faster response times and better efficiency.

Researchers at Japan's National Institute for Materials Science have reviewed the latest developments in research on photoactive organic field-effect transistors (OFETs)—devices that incorporate organic semiconductors, amplify weak electronic signals and either emit or receive light.

OFETs were developed to produce low-cost, large-area electronics such as printable flexible electronic devices. A lot of progress has been made in the development of light-emitting OFETs (LE-OFETs) since these first appeared in 2003. Research in this area has resulted in advances in the manufacture of novel organic photonics applications using cost-effective approaches.

Light emission efficiency and brightness of these transistors will improve, and further research may lead to the production of new display technologies.